

1 Summary

Model (Run-name): model1 (test)

Iteration: 5

TAC: 146153.30435161333

Took: 0.8556623458862305s

Total Time: 2.0537588596343994s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

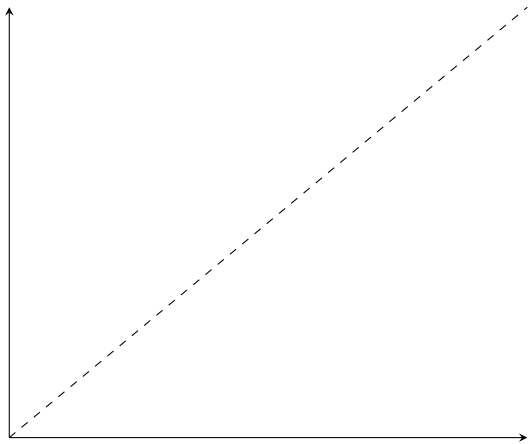
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

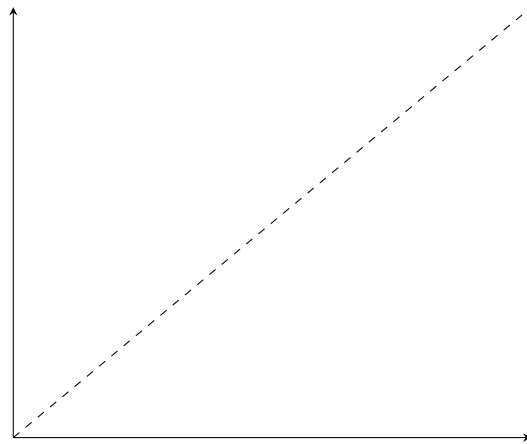
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$f_{2,1,2}^H$	5.1825831961916355	$t_{1,1,2}^H$	418.8196134269507
$z_{1,1,2}$	1.0	$f_{2,2,2}^H$	14.817416803808364	$t_{2,1,2}^H$	430.2167756716432
$z_{2,1,2}$	1.0	$f_{1,1,1}^C$	15.0	$t_{2,2,2}^H$	459.46103212427704
$z_{2,2,2}$	1.0	$f_{1,1,2}^C$	10.249298736824336	$t_{1,1,1}^C$	617.507387405553
$z_{cu,1}$	1.0	$f_{2,1,2}^C$	4.750701263175664	$t_{1,1,2}^C$	531.5414025088879
$z_{cu,2}$	1.0	$f_{2,2,2}^C$	13.0	$t_{2,1,2}^C$	650.0
$z_{hu,1}$	1.0	$RecLMTD_{1,1,1}$	0.05226976373335799	$t_{2,2,2}^C$	500.0
$A_{1,1,1}^\beta$	63.3468696286352	$RecLMTD_{1,1,2}$	0.025043680890012077	$t_{1,1,1}^{H,in}$	1566.4498056836055
$A_{1,1,2}^\beta$	10.813867750736755	$RecLMTD_{2,1,2}$	0.043252231918121886	$t_{1,1,2}^{H,in}$	4915.116162588174
$A_{2,1,2}^\beta$	79.83702775273511	$RecLMTD_{2,2,2}$	0.01134855821562564	$t_{2,1,2}^{H,in}$	3057.724085753065
$A_{2,2,2}^\beta$	44.259377040939995	$RecLMTD_{cu,1}$	0.010004362008003333	$t_{2,2,2}^{H,in}$	8742.27591424691
$A_{cu,1}^\beta$	14.535545040173215	$RecLMTD_{cu,2}$	0.010965325593640746	$t_{1,1,1}^{H,out}$	891.6714285714286
$A_{cu,2}^\beta$	18.023935870161154	$RecLMTD_{hu,1}$	0.022585152101601474	$t_{1,1,2}^{H,out}$	3617.452032499764
$A_{hu,1}^\beta$	13.192252946941991	$q_{1,1,1}$	674.778377832954	$t_{2,1,2}^{H,out}$	1917.5557825909052
$A_{1,1,1}$	63.3468696286352	$q_{1,1,2}$	1297.6641300884103	$t_{2,2,2}^{H,out}$	6792.275914246909
$A_{1,1,2}$	10.813867750736755	$q_{2,1,2}$	1140.1683031621599	$t_{1,1,1}^{C,in}$	8587.832433250569
$A_{2,1,2}$	79.83702775273511	$q_{2,2,2}$	1950.0	$t_{1,1,2}^{C,in}$	4202.212482097978
$A_{2,2,2}$	44.259377040939995	$q_{cu,1}$	827.5574920786348	$t_{2,1,2}^{C,in}$	1947.787517902022
$A_{cu,1}$	14.535545040173215	$q_{cu,2}$	1309.8316968378144	$t_{2,2,2}^{C,in}$	4550.0
$A_{cu,2}$	18.023935870161154	$q_{hu,1}$	487.3891889164761	$t_{1,1,1}^{C,out}$	9262.610811083523
$A_{hu,1}$	13.192252946941991	$t_{1,1}^{(H)}$	650.0	$t_{1,1,2}^{C,out}$	5499.876612186386
$dt_{1,1,1}$	32.492612594431996	$t_{1,2}^{(H)}$	582.5221622167046	$t_{2,1,2}^{C,out}$	3087.955821064182
$dt_{1,1,2}$	10.0	$t_{1,3}^{(H)}$	452.75574920786346	$t_{2,2,2}^{C,out}$	6500.0
$dt_{1,1,3}$	42.75574920786349	$t_{2,1}^{(H)}$	590.0		
$dt_{2,1,2}$	17.47783778329502	$t_{2,2}^{(H)}$	590.0		
$dt_{2,1,3}$	25.490489261331078	$t_{2,3}^{(H)}$	435.4915848418907		
$dt_{2,2,2}$	90.0	$t_{1,1}^{(C)}$	617.5073874055682		
$dt_{2,2,3}$	85.49158484189098	$t_{1,2}^{(C)}$	572.5221622167046		
$dt_{cu,1}$	132.7557492078635	$t_{1,3}^{(C)}$	410.0		
$dt_{cu,2}$	115.49158484189073	$t_{2,1}^{(C)}$	500.0		
$dt_{hu,1}$	62.49261259443174	$t_{2,2}^{(C)}$	500.0		
$f_{1,1,1}^H$	2.409922777974778	$t_{2,3}^{(C)}$	350.0		
$f_{1,1,2}^H$	8.457448373595133	$t_{1,1,1}^H$	504.30895275685174		

3 *RecLMTD* Tangent Points

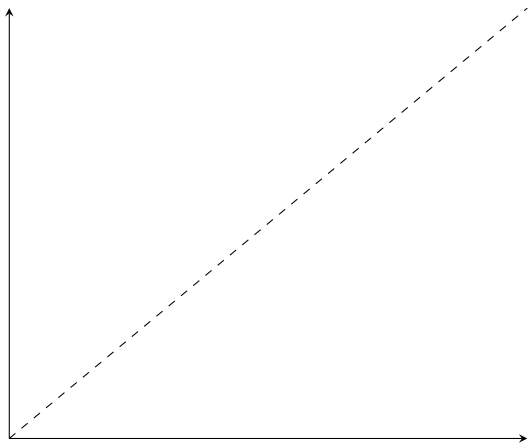
- $(1, 1, 1)$
 - Value: 0.052270
 - Error: 0.000122000930255
 - Relative error: 0.002328628001708
- $(1, 1, 2)$
 - Value: 0.025044
 - Error: 0.019312458501609
 - Relative error: 0.435395387571921
- $(2, 1, 2)$
 - Value: 0.043252
 - Error: 0.003844755579185
 - Relative error: 0.081634851473352
- $(2, 2, 2)$
 - Value: 0.011349
 - Error: 0.000050507826646
 - Relative error: 0.004430874113583
- $cu, 1$
 - Value: 0.010004
 - Error: 0.000194157390575
 - Relative error: 0.019037801761898
- $cu, 2$
 - Value: 0.010965
 - Error: 0.000041158978982
 - Relative error: 0.003739520889710
- $hu, 1$
 - Value: 0.022585
 - Error: 0.000000011448186
 - Relative error: 0.000000506889654



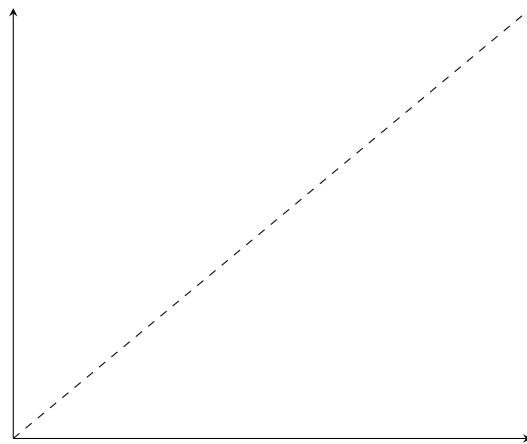
(a) *RecLMTD* - (1, 1, 1)



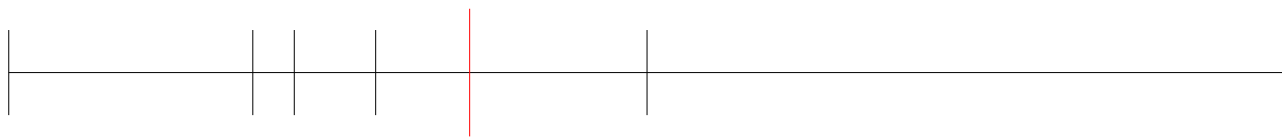
(b) *RecLMTD* - (1, 1, 2)



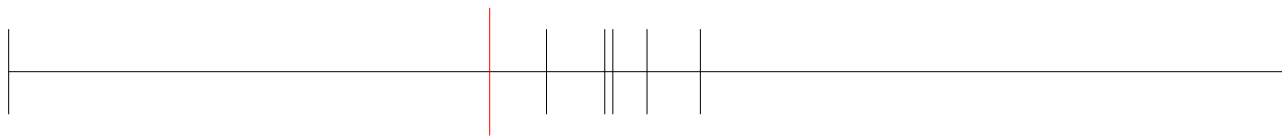
(c) *RecLMTD* - (2, 1, 2)



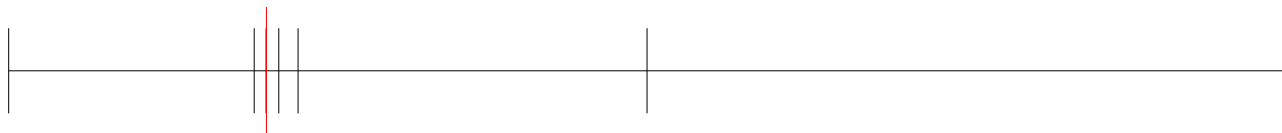
(d) *RecLMTD* - (2, 2, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1566.4498056836055
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 1, 2)
 - Value: 4915.116162588174
 - Absolute error: 11.534950834614392
 - Relative error: -0.002341337060216649
- (2, 1, 2)
 - Value: 3057.724085753065
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 8742.27591424691
 - Absolute error: 2.546585164964199e-11
 - Relative error: -2.912954463967594e-15

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 891.6714285714286
 - Absolute error: 323.6742038139146
 - Relative error: -0.26632276052915366
- (1, 1, 2)
 - Value: 3617.452032499764
 - Absolute error: 75.30677409225746
 - Relative error: 0.021260216224479232
- (2, 1, 2)
 - Value: 1917.5557825909052
 - Absolute error: 312.07844972469934
 - Relative error: -0.13996845096901286
- (2, 2, 2)
 - Value: 6792.275914246909
 - Absolute error: 15.749703846487137
 - Relative error: -0.002313402553102889

4.3 $b^{C,in}$

- (1, 1, 1)
 - Value: 8587.832433250569
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 1, 2)
 - Value: 4202.212482097978
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 1947.787517902022
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9262.610811083523
 - Absolute error: 2.2737367544323206e-10
 - Relative error: 2.4547471558576678e-14
- (1, 1, 2)
 - Value: 5499.876612186386
 - Absolute error: 51.94998688220494
 - Relative error: 0.00953573541921636
- (2, 1, 2)
 - Value: 3087.955821064182
 - Absolute error: 9.094947017729282e-13
 - Relative error: 2.945297000588225e-16
- (2, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0



(a) $t^{(H)} - (1, 1)$



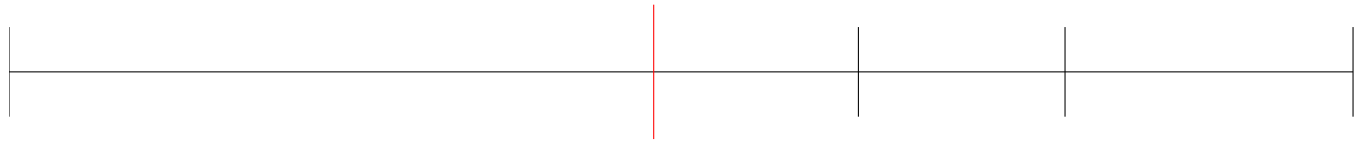
(a) $t^{(H)} - (1, 2)$



(a) $t^{(H)} - (2, 2)$



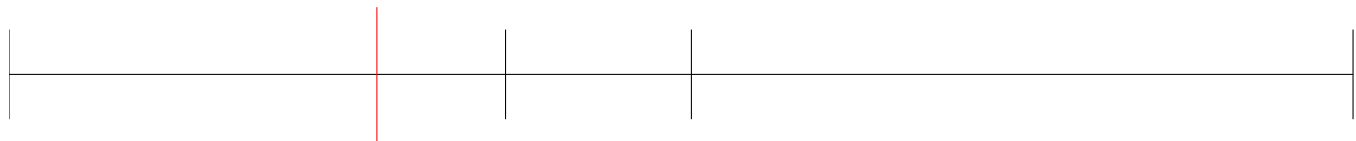
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 1, 2)$



(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(H)} - (2, 2, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (1, 3)$



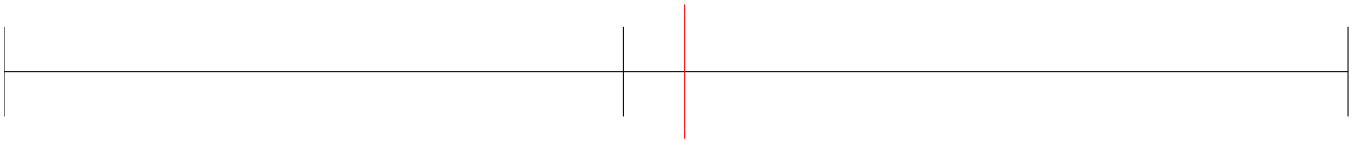
(a) $t^{(C)} - (1, 3)$



(a) $t^{(C)} - (2, 3)$



(a) $t^{(C)} - (1, 1, 1)$



(a) $t^{(C)} - (1, 1, 2)$



(a) $t^{(C)} - (2, 1, 2)$



(a) $t^{(C)} - (2, 2, 2)$

5 Area Breakpoints

5.1 A

- $(1, 1, 1)$
 - Value: 63.346870
 - Error: 7.194143134778962
 - Relative error: 0.101985254434994
- $(1, 1, 2)$
 - Value: 10.813868
 - Error: 54.182705001961772
 - Relative error: 0.833624031348825
- $(2, 1, 2)$
 - Value: 79.837028
 - Error: 18.792619995387369
 - Relative error: 0.190537231192181
- $(2, 2, 2)$
 - Value: 44.259377
 - Error: 0.000000000000000
 - Relative error: 0.000000000000000

5.2 A_{CU}

- $cu, 1$
 - Value: 14.535545
 - Error: 2.022824426206814
 - Relative error: 0.122163261927084
- $cu, 2$
 - Value: 18.023936
 - Error: 10.701526187233991
 - Relative error: 0.372544962578904

5.3 A_{HU}

- $hu, 1$
 - Value: 13.192253
 - Error: 0.017057810283752
 - Relative error: 0.001291347489453



(a) $A^\beta - (1, 1, 1)$



(a) $A^\beta - (1, 1, 2)$



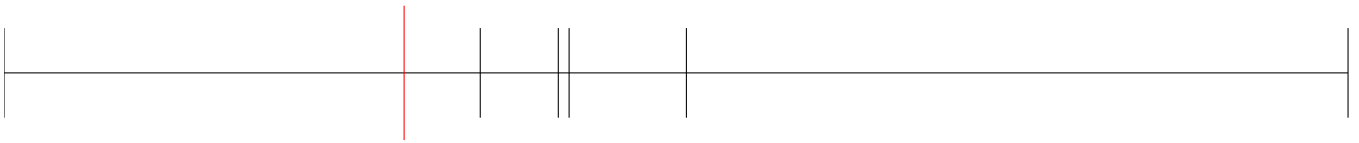
(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - (2, 2, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

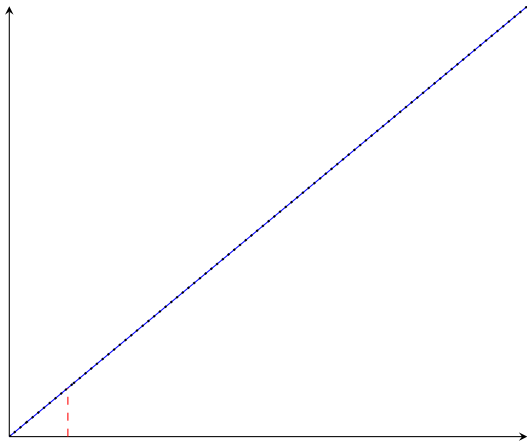


(a) $A^\beta - hu, 1$

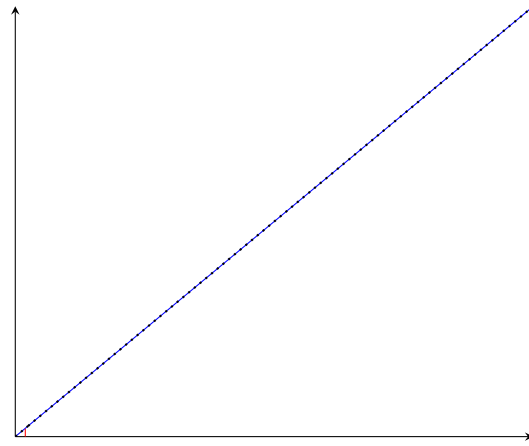
6 A^β Breakpoints

- (1,1,1)
 - q : 63.346870
 - $\hat{q}^\beta$: 63.346870
 - q^β : 63.346870
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 69.14107665413653, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1,1,2)
 - q : 10.813868
 - $\hat{q}^\beta$: 10.813868
 - q^β : 10.813868
 - Breakpoints: [0, 13.825772197515043, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2,1,2)
 - q : 79.837028
 - $\hat{q}^\beta$: 79.837028
 - q^β : 79.837028
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2,2,2)
 - q : 44.259377
 - $\hat{q}^\beta$: 44.259377
 - q^β : 44.259377
 - Breakpoints: [0, 2.171589371244973, 38.943261778500734, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 14.535545
 - $\hat{q}^\beta$: 14.535545
 - q^β : 14.535545
 - Breakpoints: [0, 2.9819181007907165, 7.702824927089129, 9.28003023700696, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$

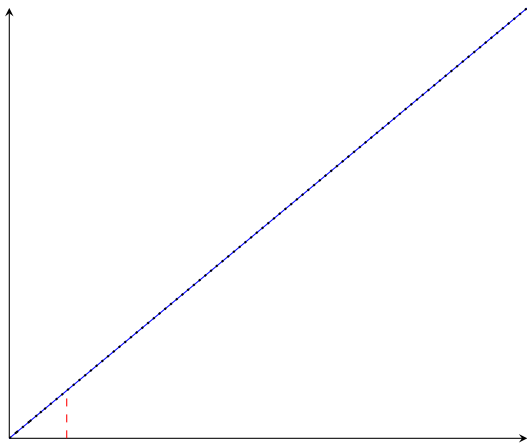
- q : 18.023936
- $\hat{q}^\beta$: 18.023936
- q^β : 18.023936
- Breakpoints: [0, 21.04660840930274, 24.97364426355679, 34.23465128608717, 38.88483572219945, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.192253
 - $\hat{q}^\beta$: 13.192253
 - q^β : 13.192253
 - Breakpoints: [0, 4.943755299006499, 12.595096067947068, 13.30871858348944, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



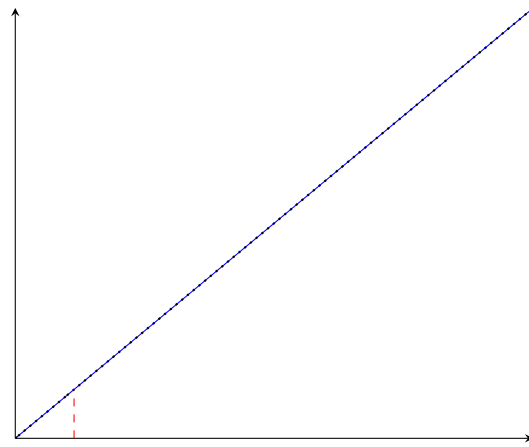
(a) $q^\beta - (1, 1, 1)$



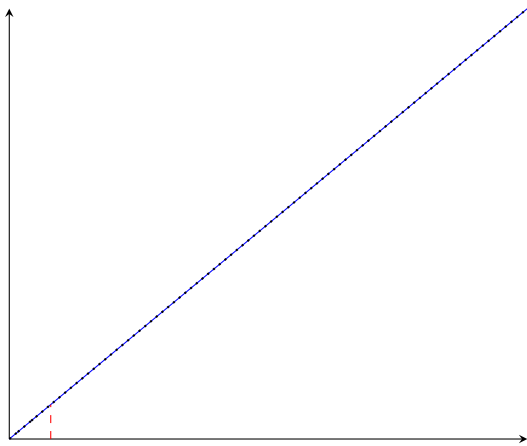
(b) $q^\beta - (1, 1, 2)$



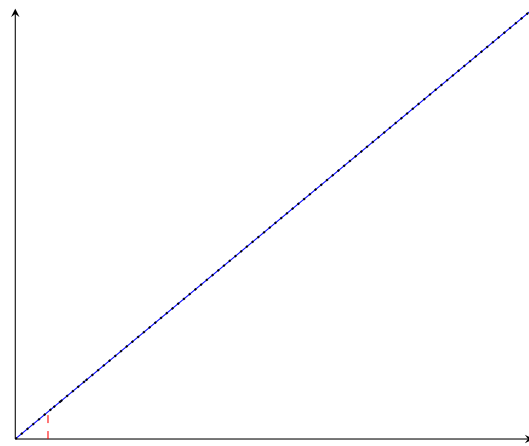
(c) $q^\beta - (2, 1, 2)$



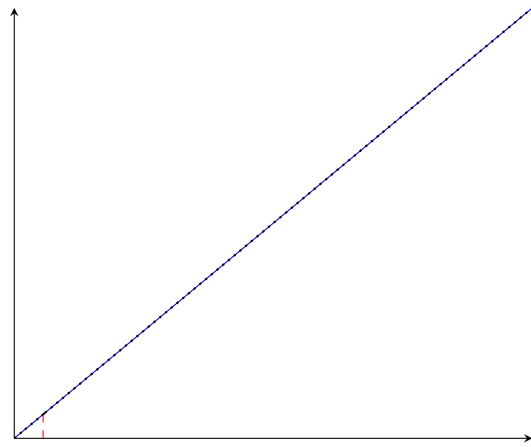
(d) $q^\beta - (2, 2, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



(a) $q^\beta - hu, 1$

7 Other variables

Variable	Value	Variable	Value
$z_{1,2,1}$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255
$z_{1,2,2}$	0.0	$q_{1,2,1}$	0.0
$z_{2,1,1}$	-0.0	$q_{1,2,2}$	0.0
$z_{2,2,1}$	-0.0	$q_{2,1,1}$	0.0
$z_{hu,2}$	-0.0	$q_{2,2,1}$	0.0
$A_{1,2,1}^\beta$	0.0	$q_{hu,2}$	0.0
$A_{1,2,2}^\beta$	0.0	$t_{1,2,1}^H$	650.0
$A_{2,1,1}^\beta$	0.0	$t_{1,2,2}^H$	425.07975671488566
$A_{2,2,1}^\beta$	0.0	$t_{2,1,1}^H$	590.0
$A_{hu,2}^\beta$	0.0	$t_{2,2,1}^H$	590.0
$A_{1,2,1}$	0.0	$t_{1,2,1}^C$	500.0
$A_{1,2,2}$	0.0	$t_{1,2,2}^C$	500.0
$A_{2,1,1}$	0.0	$t_{2,1,1}^C$	580.0
$A_{2,2,1}$	0.0	$t_{2,2,1}^C$	500.0
$A_{hu,2}$	0.0	$b_{1,2,1}^{H,in}$	4933.5501943163945
$dt_{1,2,1}$	150.0	$b_{1,2,2}^{H,in}$	910.1054595788719
$dt_{1,2,2}$	10.0	$b_{2,1,1}^{H,in}$	11800.0
$dt_{1,2,3}$	20.0	$b_{2,2,1}^{H,in}$	0.0
$dt_{2,1,1}$	24.166666666666668	$b_{1,2,1}^{H,out}$	4933.5501943163945
$dt_{2,2,1}$	220.0	$b_{1,2,2}^{H,out}$	910.1054595787793
$dt_{hu,2}$	180.0	$b_{2,1,1}^{H,out}$	11800.0
$f_{1,2,1}^H$	7.590077222025222	$b_{2,2,1}^{H,out}$	0.0
$f_{1,2,2}^H$	1.542551626404867	$b_{1,2,1}^{C,in}$	0.0
$f_{2,1,1}^H$	20.0	$b_{1,2,2}^{C,in}$	0.0
$f_{2,2,1}^H$	0.0	$b_{2,1,1}^{C,in}$	0.0
$f_{1,2,1}^C$	0.0	$b_{2,2,1}^{C,in}$	6500.0
$f_{1,2,2}^C$	0.0	$b_{1,2,1}^{C,out}$	0.0
$f_{2,1,1}^C$	0.0	$b_{1,2,2}^{C,out}$	0.0
$f_{2,2,1}^C$	13.0	$b_{2,1,1}^{C,out}$	0.0
$RecLMTD_{1,2,1}$	0.01607047381606835	$b_{2,2,1}^{C,out}$	6500.0
$RecLMTD_{1,2,2}$	0.1		
$RecLMTD_{2,1,1}$	0.1		
$RecLMTD_{2,2,1}$	0.02011172883803818		